Customer Review Intelligence Analysis for E-commerce Platform

Goal

Identify customer sentiment, major complaints, and operational improvements from large-scale review data.

1. Problem

Businesses today receive a very large volume of customer reviews across different platforms. However, most of this feedback is not used effectively. Companies often read reviews individually, but they do not have a structured way to analyse thousands of comments at once.

This creates a gap between customer feedback and business decision making.

The purpose of this project is to analyse large-scale customer review data and transform it into clear and actionable insights. The focus is to identify sentiment patterns, key complaint areas, and opportunities for improving operations and customer experience.

2. Dataset

The dataset used in this project consists of customer reviews from Shopee, a large e-commerce platform.

- Total reviews: 782,079
- Columns included:
    - review_text
    - rating
    - author_id
    - author_name

This dataset represents a realistic example of how businesses receive customer feedback at scale. It includes a mix of positive, neutral, and negative reviews, which makes it suitable for sentiment and theme analysis.

Large datasets like this are important because they provide a more reliable view of customer behaviour compared to small samples.

3. Methodology

The analysis was conducted using a structured data analytics pipeline. The goal was to move from raw text data to business insights.

The main steps are described below.

Data Cleaning and Preparation

The review text was first cleaned to remove unnecessary characters, duplicates, and formatting issues. This step ensures that the analysis is consistent and reliable.

Sentiment Analysis

Each review was analysed to determine whether it is positive, neutral, or negative. This helps to understand the overall customer perception of the platform.

Theme Detection

Keyword-based clustering was used to group reviews into common themes. This allows the identification of recurring topics such as service quality, product issues, or user experience.

Identification of Drivers

Both positive drivers (what customers like) and negative drivers (what customers complain about) were identified. This step is important for linking insights to business actions.

Insight Generation

Finally, the results were translated into business insights and recommendations. The objective is not only to analyse data, but to support decision making.

4. Analytics Tools

The project was developed using the following tools and technologies:

- Python
- pandas for data manipulation
- NLP techniques for sentiment scoring
- Data visualisation for result interpretation
- Automated report generation

These tools were combined into a workflow that allows the analysis to be repeated with different datasets.

5. Key Insights

The analysis produced several important findings.

From the sentiment distribution, positive reviews accounted for approximately 28.9% of all feedback, while negative reviews represented a smaller proportion. Many reviews were neutral, which indicates that many customers are satisfied but not strongly engaged.

The most frequently mentioned themes were:

- Product and Service Quality
- Service Quality
- Location and Accessibility

Product and service quality appeared as the main strength of the platform. Many customers mentioned convenience, variety of products, and ease of use.

However, service quality was identified as the main area for improvement. Some reviews highlighted issues such as slow response times, unhelpful support, and difficulty resolving problems.

This combination of strong product offering and weaker service experience suggests that improvements in customer support could significantly increase overall satisfaction.

6. Business Impact

The insights from this analysis can directly support business decision making.

If service quality issues are addressed, the platform could improve customer satisfaction and increase repeat purchases. Customer support is often a key factor in user retention, especially in competitive e-commerce environments.

In addition, review analysis can help businesses:

- Improve operational processes by identifying recurring issues
- Strengthen marketing messaging by focusing on what customers value
- Detect product or seller issues earlier
- Reduce negative feedback before it impacts brand reputation

This type of analysis allows companies to move from reactive decision making to a more proactive and data-driven approach.

7. Sample Dashboard and Report

To support the analysis, an automated review intelligence tool was developed. The tool allows users to upload review data, preview the dataset, and generate insights.

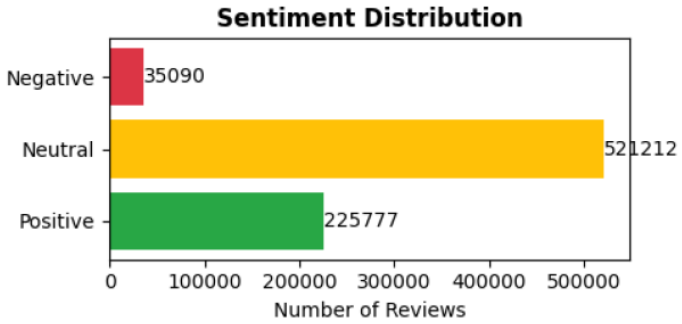
The system includes:

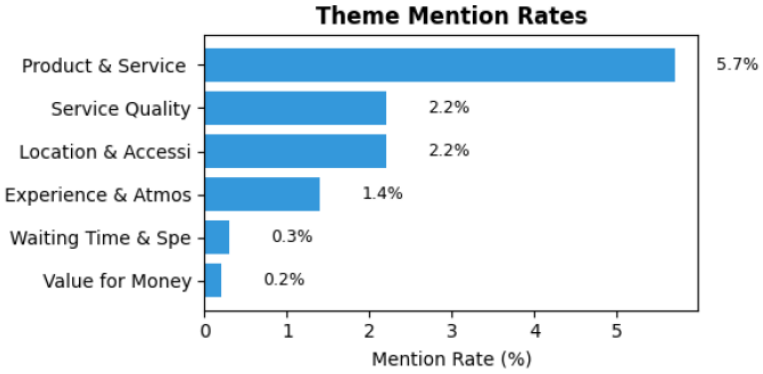
- Data upload interface
- Review preview table
- Automated sentiment analysis
- Theme detection
- Report generation

The output is a structured report that includes sentiment distribution, key themes, strengths, risks, and recommendations.

This approach demonstrates how analytics can be transformed into a practical business tool rather than a one-time analysis.

Sentiment & Theme Analysis





Theme Details

Theme	Mention Rate	Impact Score	Business Impact
Product & Service Qu	5.7%	3.6	High
Service Quality	2.2%	1.1	High
Location & Accessibi	2.2%	0.6	Medium
Experience & Atmosph	1.4%	-0.1	Medium
Waiting Time & Speed	0.3%	-0.0	High
Value for Money	0.2%	0.0	High

8. Recommendations

Based on the findings, several recommendations can be made.

Operational Improvements

- Improve customer support responsiveness to reduce frustration
- Implement stronger seller verification to reduce defective or low-quality products
- Create clear service recovery processes for handling complaints
- Monitor recurring issues through continuous review analysis

Marketing Improvements

- Highlight product quality and reliability in marketing campaigns
- Use customer testimonials to build trust
- Emphasise convenience and ease of use
- Promote positive customer experiences across channels

By combining operational improvements with targeted marketing, the business can improve both customer satisfaction and brand perception.

9. Conclusion

This project shows how large-scale customer review data can be transformed into meaningful business insights.

Instead of treating reviews as isolated comments, this approach allows companies to identify patterns, understand customer needs, and act based on data.

The development of an automated review intelligence tool also demonstrates the potential to scale this type of analysis across different businesses and industries.

Overall, the project highlights the value of combining data analytics with business understanding to support better decision making.